3.3.9 Carboxylic acids and derivatives (A-level only)

Carboxylic acids are weak acids but strong enough to liberate carbon dioxide from carbonates. Esters occur naturally in vegetable oils and animal fats. Important products obtained from esters include biodiesel, soap and glycerol.

3.3.9.1 Carboxylic acids and esters (A-level only)

Content	Opportunities for skills development
The structures of: • carboxylic acids • esters. Carboxylic acids are weak acids but will liberate CO_2 from carbonates. Carboxylic acids and alcohols react, in the presence of an acid catalyst, to give esters. Common uses of esters (eg in solvents, plasticisers, perfumes and food flavourings). Vegetable oils and animal fats are esters of propane-1,2,3-triol (glycerol). Esters can be hydrolysed in acid or alkaline conditions to form alcohols and carboxylic acids or salts of carboxylic acids. Vegetable oils and animal fats can be hydrolysed in alkaline conditions to give soap (salts of long-chain carboxylic acids) and glycerol. Biodiesel is a mixture of methyl esters of long-chain carboxylic acids. Biodiesel is produced by reacting vegetable oils with methanol in the presence of a catalyst.	**AT b, d, g and k** **PS 4.1** Students could make esters by reacting alcohols with carboxylic acids, purifying the product using a separating funnel and by distillation. **AT b, d, g, h and k** Students could identify an ester by measuring its boiling point, followed by hydrolysis to form the carboxylic acid, which is purified by recrystallisation, and determine its melting point. **AT b, c, d and k** Students could make soap. **AT b and k** Students could make biodiesel.

3.3.9.2 Acylation (A-level only)

<table>
<tr><th>Content</th><th>Opportunities for skills development</th></tr>
<tr><td>The structures of:
• acid anhydrides
• acyl chlorides
• amides.

The nucleophilic addition–elimination reactions of water, alcohols, ammonia and primary amines with acyl chlorides and acid anhydrides.

The industrial advantages of ethanoic anhydride over ethanoyl chloride in the manufacture of the drug aspirin.

Students should be able to outline the mechanism of nucleophilic addition–elimination reactions of acyl chlorides with water, alcohols, ammonia and primary amines.</td><td>AT d and k

PS 2.2

Students could record observations from reaction of ethanoyl chloride and ethanoic anhydride with water, ethanol, ammonia and phenylamine.

AT b, d, g and h

PS 2.1, 2.3 and 4.1

Students could carry out the preparation of aspirin, purification by recrystallisation and determination of its melting point.

Students could carry out the purification of impure benzoic acid and determination of its melting point.</td></tr>
<tr><td>Required practical 10

Preparation of:
• a pure organic solid and test of its purity
• a pure organic liquid.</td><td></td></tr>
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